

TED (15) 4023
RIVISION 2015

Reg. No:.....
Signature

FOURTH SEMESTER DIPLOMA EXAMINATION IN MECHANICAL ENGINEERING/TECHNOLOGY

MODEL QUESTION PAPER METTALLURGY AND MACHINE TOOLS

(Maximum marks :100)

(Duration : 3 hrs)

PART-A (Maximum marks :10)

I.(Answer the following questions in one or two sentences .Each question carry two marks)

- 1) List any four crystal defects
- 2) Name various types of chips
- 3) Define machinability
- 4) What is the use of drill bit
- 5) Define gang milling

PART B

Maximum marks: 30

II. (Answer any five of the following questions. Each question carries 6 marks)

- 1) Draw the cooling curve for pure iron
- 2) List the advantages of powder metallurgy
- 3) Explain desirable properties of a cutting fluid
- 4) Briefly explain any three lathe operations
- 5) Classify the drilling machine
- 6) Sketch the nomenclature of a plain milling cutter
- 7) Explain functions of a shaper

PART C

(Maximum marks : 60)

(Answer one full question from each unit. Each question carries 15 marks)

MODULE-I

- III. (a) draw iron carbon equilibrium diagram and indicate different phases 8
(b) Describe the continuous cooling transformation (CCT) diagram 7

OR

- IV. (a) List various heat treatment process and explain any two 8
(b) Describe any two methods for making metal powders 7

MODULE-II

- V. (a) Differentiate between orthogonal and oblique cutting 8
(b) Sketch the nomenclature of a single point cutting tool 7

OR

- VI. (a) Sketch the line diagram of a Centre lathe and mark important parts 8
(b) List any four taper turning methods. Explain any two 7

MODULE-III

- VII. (a) Draw the line diagram of a Drilling machine and mark the parts 8
(b) Show the nomenclature of a taper shank twist drill 7

OR

- VIII. (a) Differentiate up milling and down milling 8
(b) What is indexing. Explain simple indexing in milling machine 7

MODULE-IV

- IX. (a) Explain whitworth quick return mechanism in shaper 8
(b) Describe uses of slotter 7

OR

- X. (a) Compare shaper, slotter and planer 8
(b) List the specification of a planner 7